Den box internal probes

t-entrance

Thermistor probe-104K

th-01

gnd

t-den

Thermistor probe-104K

th-02

gnd

Li-AAA-2

B2

Batt+Ve

batt-Ve

Under box enclosure

PROBE-HEADER
U10

R4
18k

R5
18k

U5
XIAO NRF52840

P0.02_A0_D0
P0.03_A1_D1
P0.28_A2_D2
P0.29_A3_D3
P0.04_A4_D4_SDA
P0.05_A5_D5_SCL
P1.11_D6_TX
P1.12_D7_RX
P1.13_D8_SCK
P1.14_D9_MISO
P1.15_D10_MOSI

P0.10_NFC2
P0.09_NFC1
BAT-
BAT+
GND
RESET
SWCLK
SWDIO
VBUS
GND
3V3

Q2

Battery Rev-Chrg Protect SI2305CDS-T1-BE3

Batt+Ve

R3
1k

D1
BZX84-C5V1

R2
100k

batt-Ve

CN1

Batt+Ve

batt-Ve

2.0-2P ZZDK-R PCB CONNECTOR

U11
SHT41-AD1F-R2

SDA VSS
SCL VDD

C5
0.1u

R1
10k

R6
10k

Logging:

Every 30-seconds wake from power save and measure:

Den probe temperature t-den degC

Entrance probe difference: ent-delta = (t-den MINUS t-ent)

Save data in RAM to store during EEPROM write cycle

User Interface:

Every 1 minute BLE beacon:

Smartphone connectivity

GUI for evaluation, download and config

TITLE: Den Box Logger - Seed XIAO prototype		REV: 2.0
EasyEDA	Company: Den-Box-Logger	Sheet: 1/3
	Date: 2026-11-08 Drawn By: Aidan Hedley	